

Unit 2 - Gauss's law

Flux through closed surface = Q enclosed / ϵ_0

Choose a surface with symmetry:

sphere $\rightarrow$ point charge, $E = kQ/r^2$

cylinder $\rightarrow$ line charge, $E = \lambda/(2\pi\epsilon_0 r)$

pillbox $\rightarrow$ sheet, $E = \sigma/(2\epsilon_0)$

Field lines of a point charge

